Parsa Oryani

M.Sc. Computer Engineering – Secure Computing — Tehran, Iran

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Research Profile

M.Sc. student in Secure Computing at Sharif University of Technology. Current research focuses on **security of LLM-based agentic and multi-agent systems**, particularly information-flow control, authorization, provenance, secure memory/tool interactions, and runtime enforcement. Background in deep learning, cryptography, distributed systems, blockchain security, and formal security analysis.

Education

Sharif University of Technology — Sep 2025 – Present
M.Sc. in Computer Engineering, Secure Computing — Tehran, Iran

- National M.Sc. Computer Engineering Entrance Examination: **Rank 16** among approximately 18,000 participants.
- Selected coursework: Applied Cryptography (18.9/20), Deep Learning (18.6/20), Secure Software Systems (18.2/20), Formal Methods in Information Security, and Foundations and Applications of Blockchain.
- Research interests: LLM-agent security, information-flow control, authorization, provenance, secure agent architectures, and formal methods.

Amirkabir University of Technology (Tehran Polytechnic) — Sep 2020 – Feb 2025
B.Sc. in Computer Science — Tehran, Iran

- National University Entrance Examination (Mathematics & Technical Sciences): **Rank 343 among 135,436 participants**.
- Cryptography I (18.75/20); Special Topics in Cryptography (18.25/20, highest grade in class).

Research Experience

Research Assistant – Security of LLM-Based Agentic and Multi-Agent Systems — Sep 2025 – Present
Sharif University of Technology; Supervisor: Dr. Morteza Amini — Tehran, Iran

- Investigating security and privacy of LLM-based agentic and multi-agent systems, including information-flow control, authorization, provenance, memory/state, tool/skill security, and runtime enforcement.
- Studying trust boundaries, least privilege, secure composition, update safety, and recovery from unsafe effects through systems security, program analysis, formal methods, and privacy.
- Earlier research focused on blockchain security, especially cross-chain and Layer-2 interoperability: state verification, bridge trust assumptions, finality, atomic execution, and protocol safety/liveness.

Teaching Experience

Computer Networks — *Undergraduate* — Feb – Jun 2026

Sharif Univ. of Technology | Instructor: Dr. Sadeghzadeh — Designed and graded theoretical and practical assignments.

Machine Learning — *Undergraduate* — Feb – Jun 2026

Sharif Univ. of Technology | Instructor: Dr. Motahari — Designed theoretical assignments and evaluated student submissions.

Artificial Intelligence & Lab — *Undergraduate* — Feb – Jun 2025

Amirkabir Univ. of Technology | Instructor: Dr. Ghatee — Designed and graded theoretical and practical assignments.

Design & Analysis of Algorithms — *Undergraduate* — Oct 2023 – Jan 2024

Amirkabir Univ. of Technology | Instructor: Dr. Ardeshir Dolati — Designed and graded theoretical and practical assignments.

Foundations of Matrices & Linear Algebra — *Undergraduate* — Sep 2023 – Jan 2024

Amirkabir Univ. of Technology | Instructor: Dr. Najafi — Designed assignments and quizzes; graded coursework and led problem-solving sessions.

Advanced Programming — *Undergraduate* — Feb – Jun 2023

Amirkabir Univ. of Technology | Instructor: Dr. Bejani — Designed programming assignments and projects; conducted workshops and review sessions.

Technical Skills

AI & Agent Systems	PyTorch, Transformers, RAG, MCP/tool integration, agent orchestration, model evaluation.
Security & Verification	Information-flow control, taint/data-flow analysis, authorization policies, CodeQL, Z3/SMT, model checking.
Blockchain Protocols	Solidity, EVM, Ethereum JSON-RPC, Hyperledger Besu, QBFT, bridge protocols, cross-chain state verification.
Software & Systems	Python, Rust, TypeScript, C++, PostgreSQL, Docker, Linux, Git, FastAPI, Prometheus, Grafana.